OphidianAI

PERFORMANCE ROADMAP

Internal Project Roadmap

March 9, 2026

Version 2.0

INTELLIGENCE, ENGINEERED.

Roadmap Overview

This roadmap prioritizes performance, stability, and code quality before adding new features or services. The goal is a lean, fast, production-hardened website that serves as both a business platform and a portfolio piece.

CURRENT SCORE

Phase 1: Critical Fixes	**2 items**
Phase 2: Performance	**4 items**
Phase 3: Code Quality	**4 items**
Phase 4: Feature Expansion	**4 items**
Total Items	**14**
Target Score	**98%+**

Execution Order

Each phase must be completed before moving to the next. Performance and stability come first, features come last.

Phase 1: Critical Fixes (P0)
Launch blockers. Must be resolved before the site can go live on the custom domain. Estimated: 20 minutes.

Phase 2: Performance Optimization (P1)
Bundle size, lazy loading, error resilience. Reduces initial page load by ~200KB+. Estimated: 2-3 hours.

Phase 3: Code Quality (P2-P3)
Caching, component splitting, dependency cleanup, dead code removal. Estimated: 3-4 hours.

Phase 4: Feature Expansion (P3-P4)
Stripe checkout, blog content, Calendly, legal pages. Only after the foundation is solid. Estimated: 8-12 hours.

Phase 1: Critical Fixes

Launch blockers -- resolve before going live

1.1 Connect ophidianai.com custom domain DONE

Cloudflare DNS proxy was blocking Vercel's http-01 challenge. Resolved by setting DNS to "DNS only" (gray cloud) for SSL cert generation. Domain is now connected.

1.2 Set RESEND_API_KEY in Vercel P0

Impact: Contact form email delivery fails silently in production.
Fix: Create Resend account, get API key, add as env var in Vercel project settings.
Effort: 5 minutes.
Files: Vercel dashboard only (no code changes).

Phase 1 Status

1 of 2 items complete. Remaining item is a configuration task (no code changes required).

Phase 2: Performance

Bundle optimization and error resilience

2.1 Lazy-load Three.js and Glass Orb P1

Impact: Three.js (~183KB gzipped) loads on every page via the AI chat widget's GlassOrb3D component, even before the user opens chat.
Fix: Wrap GlassOrb3D import with next/dynamic({ ssr: false }). Only load when chat is opened or orb is visible.
Effort: 30 minutes.
Files: src/components/ui/ai-orb.tsx, possibly a new wrapper component.

2.2 Add error boundaries for 3D/WebGL components P1

Impact: If WebGL fails (unsupported device, GPU crash), the entire page crashes with no recovery.
Fix: Wrap GlassOrb3D, SiriOrb, and any Canvas components in React error boundaries with graceful fallbacks.
Effort: 30 minutes.
Files: New src/components/ui/webgl-error-boundary.tsx, modify ai-orb.tsx.

2.3 Remove duplicate motion package P1

Impact: Both "framer-motion" and "motion" are installed. "motion" is the modern successor package. Having both adds unnecessary bundle weight.
Fix: Audit all imports, consolidate to "motion" (the newer package), remove "framer-motion" from package.json.
Effort: 30 minutes.
Files: package.json, any file importing from "framer-motion".

2.4 Optimize content queries with caching P1

Impact: usePageContent fetches from Supabase on every page navigation with no cache. Adds latency and unnecessary DB calls.
Fix: Add client-side cache (React context or simple Map) that persists across navigation. Invalidate on save.
Effort: 45 minutes.
Files: src/lib/use-page-content.ts, possibly src/lib/edit-mode-context.tsx.

Phase 3: Code Quality

Maintainability and architectural improvements

3.1 Split dashboard into sub-components P2

Impact: dashboard/page.tsx is 691 lines -- sidebar, stat cards, charts, activity feed, integrations all in one file.
Fix: Extract into DashboardSidebar, StatCard, UsageChart, ActivityFeed, IntegrationsList components.
Effort: 1.5 hours.
Files: src/app/dashboard/page.tsx, new files in src/app/dashboard/components/.

3.2 Remove hardcoded env fallbacks from next.config.ts P2

Impact: Supabase URL and anon key are hardcoded as fallback values in next.config.ts. Not a security risk (public keys), but poor practice -- masks missing env vars.
Fix: Remove fallback values. Let build fail if env vars are missing (fail-fast).
Effort: 15 minutes.
Files: next.config.ts.

3.3 Audit and remove unused dependencies P3

Impact: @tsparticles packages may not be actively used. @paper-design/shaders-react usage should be verified. Unused deps add install time and potential vulnerabilities.
Fix: Search codebase for each dependency's import. Remove any that are unused.
Effort: 30 minutes.
Files: package.json.

3.4 Add basic E2E test suite P3

Impact: Zero tests in the project. No safety net for regressions. High risk when making changes to auth or CMS.
Fix: Add Playwright E2E tests for: (1) public page navigation, (2) sign-in flow, (3) edit mode toggle + save, (4) contact form submission.
Effort: 4 hours.
Files: New tests/ directory, playwright.config.ts, package.json scripts.

Phase 4: Feature Expansion

New capabilities -- only after Phases 1-3 are complete

4.1 Stripe checkout integration P3

Impact: Revenue enablement. Clients can purchase subscription plans directly from the pricing page.
Fix: Stripe Checkout session creation, webhook for subscription events, pricing page "Get Started" buttons wired to checkout.
Effort: 4 hours.
Files: New API route, pricing-section.tsx, new webhook handler.

4.2 Calendly embed on contact page P3

Impact: Lead conversion. Prospects can book discovery calls directly instead of waiting for email reply.
Fix: Embed Calendly inline widget or popup on the contact page alongside the form.
Effort: 30 minutes (once Calendly account exists).
Files: src/app/contact/page.tsx.

4.3 Blog content (5+ posts) P4

Impact: SEO authority and thought leadership. Currently only 1 blog post.
Fix: Write and publish 5+ MDX posts covering AI integration topics, case studies, and industry insights.
Effort: Ongoing (1-2 posts per week).
Files: src/content/blog/*.mdx.

4.4 Privacy Policy and Terms of Service P4

Impact: Legal compliance. Required before processing payments or collecting personal data at scale.
Fix: Create /privacy and /terms pages with appropriate legal content.
Effort: 2 hours.
Files: New pages in src/app/privacy/ and src/app/terms/.

Implementation Summary

#	ITEM	PHASE	PRIORITY	EFFORT	STATUS
1.1	Connect custom domain	1	P0	15 min	DONE
1.2	Set RESEND_API_KEY	1	P0	5 min	TODO
2.1	Lazy-load Three.js / glass orb	2	P1	30 min	TODO
2.2	Error boundaries for 3D components	2	P1	30 min	TODO
2.3	Remove duplicate motion package	2	P1	30 min	TODO
2.4	Cache content queries	2	P1	45 min	TODO
3.1	Split dashboard components	3	P2	1.5 hrs	BACKLOG
3.2	Remove hardcoded env fallbacks	3	P2	15 min	BACKLOG
3.3	Audit unused dependencies	3	P3	30 min	BACKLOG
3.4	Add E2E test suite	3	P3	4 hrs	BACKLOG
4.1	Stripe checkout	4	P3	4 hrs	BACKLOG
4.2	Calendly embed	4	P3	30 min	BACKLOG
4.3	Blog content (5+ posts)	4	P4	Ongoing	BACKLOG
4.4	Privacy / Terms pages	4	P4	2 hrs	BACKLOG

Effort Summary

Phase 1: Critical Fixes
~5 minutes remaining (1 config task)

Phase 2: Performance
~2.25 hours (4 code tasks)

Phase 3: Code Quality
~6.25 hours (4 tasks)

Phase 4: Features
~7+ hours (4 tasks, blog ongoing)

Execution Plan
Start with Phase 2 performance items immediately (Phase 1 remaining item is a manual config task). Work through items sequentially within each phase. Commit after each item. Build and verify after each commit.

Definition of Done
Each item is complete when: (1) code changes are committed and pushed, (2) Vercel deployment succeeds with zero build errors, (3) the fix is verified in production. No item is marked DONE until all three criteria are met.